Quin - Motion Algorithms Engineer
Location: Hyderabad
Experience: 4 - 6 years

Job Overview:

We are looking for a highly skilled **Sensor Fusion & Motion Algorithms Engineer** with deep expertise in **Digital Signal Processing (DSP)** and real-time sensor data analysis. This role focuses on building advanced algorithms for motion tracking, impact detection, and GPS-based analytics.
You will work closely with hardware and firmware teams to design robust, real-time algorithms for IMU and GPS data, and contribute to building scalable data pipelines and analytics infrastructure.

Key Responsibilities:

- Design and implement advanced **Digital Signal Processing (DSP)** algorithms for real-time systems
- Develop **motion analysis and sensor fusion algorithms** using:
    - Accelerometers
    - Gyroscopes
    - IMUs and other motion sensors
- Analyse high-frequency, real-time sensor data for:
    - Noise reduction
    - Signal conditioning
    - Feature extraction
- Build algorithms for:
    - Motion tracking and activity classification
    - Impact detection, magnitude estimation, and direction analysis
    - Event detection
- Develop **GPS-based algorithms**, including:
    - Filtering (Kalman filters, smoothing, outlier rejection)
    - Map matching and trajectory correction
    - Distance, speed, and elevation estimation
- Implement **sensor fusion techniques** combining IMU and GPS data
- Work on **real-time and offline data processing pipelines**
- Define and build **data architecture** for:
    - Data ingestion
    - Storage
    - Retrieval
    - Processing and analytics
- Collaborate in building **dashboards and monitoring tools** for:
    - Data quality
    - Signal health
    - Anomaly detection
- Work closely with firmware and hardware teams to:
    - Ensure algorithm feasibility on embedded systems
    - Optimize performance and power consumption
    - Validate sensor data integrity

Required Skills & Qualifications:

- Strong expertise in **Digital Signal Processing (DSP)**, including:

        - Filtering (low-pass, high-pass, band-pass)
        - Time-series analysis
        - Noise reduction techniques
    - Strong experience with **IMU data processing and motion analysis**
    - Hands-on experience with **sensor fusion algorithms**:
        - Complementary filters
        - Kalman filters (EKF/UKF preferred)
    - Experience working with **GPS data processing**, including filtering and trajectory analysis
    - Strong programming skills in:
        - Python (for prototyping and analysis)
        - C/C++ (for embedded or performance-critical implementations)
    - Experience handling **real-time, high-frequency data streams**
    - Strong mathematical foundation in:
        - Linear algebra
        - Probability and statistics
        - Signal processing
    - Good understanding of **physics (mechanics, motion dynamics)**

Preferred Qualifications:

- Experience building algorithms for **wearables or IoT devices**
- Familiarity with **embedded constraints** (memory, compute, power)
- Experience with **machine learning for time-series data** (optional but valuable)
- Experience with **data pipelines and backend systems** (e.g., streaming, batch processing)
- Familiarity with **data visualization tools and dashboarding** (Grafana, custom dashboards, etc.)
- Experience with **map matching algorithms and geospatial data processing**
- Prior work on impact biomechanics or safety systems/sports analytics or fitness tracking systems

Soft Skills:

- Strong analytical and problem-solving mindset
- Ability to translate raw sensor data into meaningful insights
- Good collaboration skills across hardware, firmware, and data teams
- Strong documentation and communication skills

Cross-Functional Collaboration:

- Work closely with **Hardware Engineers** to understand sensor characteristics and limitations
- Collaborate with **Firmware Engineers** to:
    - Optimize algorithms for embedded deployment
    - Ensure real-time performance and efficiency
- Partner with **Data/Backend teams** to build scalable analytics pipelines and dashboards

Key Priority Requirements

1. Expertise in Motion data analysis - GPS, Accelerometers, Gyroscopes, etc.
2. Expertise in Digital Signal Processing - Filtering on data
3. Expertise in Algo development on Motion data, and Data Architecture design